22	A	E = W/Q = (2500+1500) / 2000 = 2 V
23	A	Effective resistance across the 6Ω, 10Ω and 4Ω resistors = $[\left(\frac{1}{6}\right)+\left(\frac{1}{10}\right)+\left(\frac{1}{4}\right)]^{-1}=1.935\Omega$ By potential divider rule, voltage across the 10 Ω resistor = (12.0)(1.935)/(2+1.935) = 5.9V Current through the 10 Ω resistor = 5.9/10 = 0.59 A

24	B	Using right hand grip rule, the magnetic field on Y due to X and Z is into the plane of paper. Using Fleming's Left-Hand Rule, both X and Z will cause a magnetic force on Y that points towards X.
25	B	Use Fleming's Right-Hand Rule, the induced current will flow from O outwards to the rim.
26	D	Secondary r.m.s voltage = $\frac{1000}{200}20 = 100V$ Secondary r.m.s current = $\frac{100}{100} = 1.00A$ Primary r.m.s. current = $\frac{1000}{200}1.00 = 5.00A$ Primary peak current = $5 \times \sqrt{2} = 7.07V$
27	C	A: A higher work function should lead to a photoelectron with lower kinetic energy. B: Stopping potential is not a energy quantity. C: Once an electron at the metal surface absorbs a photon with energy higher than the work function of the metal it will be emitted immediately. D: Emission of a photoelectron only depends on the energy of the incident photon.
28	B	The diagram shows that there is a range of frequencies for emitted photon, hence there is a range of energies for the emitted photon. The energy-time uncertainty principle best accounts for this phenomenon.
29	C	β-particles have a negative charge and is affected by magnetic and electric fields unlike X-rays which are photon particles. β-particles will be stopped by a few mm of Al, will X-rays will not be stopped by Al or container of cargo truck. Both have energy that can ionize air particles.
30	B	Reading off the log graph, when A/A_o = 0.5, the time pass is 160 years.